



Course: Deep Learning for Visual Computing

Final Project

Mushroom Classification

Group number: 6



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Abstract

This project investigates how well the lightweight MobileNetV2 architecture scales from simple to increasingly challenging mushroom classification tasks. Across three datasets of rising visual complexity, the model performs strongly on distinct species but worsens when species share similar colors and shapes. The more difficult datasets require reduced learning rates, stronger regularisation, and full-network fine-tuning to achieve stable training. Although the final model reached solid performance on twelve visually similar species, errors remained concentrated among classes with overlapping appearance. The study demonstrates both the potential and the limitations of lightweight CNNs for practical mushroom identification.

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1 Introduction

In recent years, automated image recognition systems have found widespread use in domains such as medicine, agriculture, and environmental monitoring. One particular interesting area is the potential for mushroom detection and classification, where deep learning offers the potential to accurately distinguish between species. This has always been a challenge for people and is essential when talking about food safety, because of the high amount of poisonous mushrooms.

Inspired by recent research in the field, particularly the 2022 study by Demirel "*Deep Learning Approach for Classification of Mushrooms*"[1], we intend to replicate his training approach. This study achieved high accuracy in identifying a limited number of visually distinct mushroom species, however, we are skeptical about its scalability. In this project, we will replicate the setup of the article as a starting point, to see if we can achieve the same level of success. From there we will try to run the same model on two more complex datasets; one with visually similar mushrooms and one with a larger number of species.

2 Related work

Most of the earlier research on visual recognition of mushrooms is conducted by focusing on adapting pre-trained CNN architectures to small mushroom datasets. For instance, we found a study from Zahan et al[2] that applied models such as Inceptionv3, VGG16 and ResNet50 to identify and distinguish between edible, inedible and poisonous mushrooms. This approach highlighted the utility transfer learning could offer in these models, but critically the models were limited by its reliance on global color and shape cues, rather than morphological distinctions. Even with these limitations, this study was able to achieve 88.4% accuracy. A similar study by Sutayco and Caya[3], that developed a CNN based recognition system for six species on a Raspberry Pi, showcased that compact models were able to achieve a high accuracy of 92.7%. With the limitation being that this study was conducted in a controlled setting, meaning the real world application accuracy might fall short.

2.1 The inspired approach

The study we chose to replicate for our project is by Demirel[1], who proposed a lightweight CNN framework based on MobileNetV2. This study enhanced their model with global average pooling and fully connected layers. With a dataset consisting of 5.470 images of four mushroom species. This study showcased that with a simple end to end training procedure and some data augmentation, their model achieved a 97.89% accuracy. By outperforming other pre-trained CNNs such as ResNet50 and VGG16, the study demonstrated that efficient architectures can achieve high accuracy even with limited data. However, the limitations of this study, where the controlled setting and balance in the dataset may lead to an overestimation of model generalization. The

scalability and robustness of MobileNetV2 under more challenging conditions, including a broader dataset and highly similar mushroom species, create a relatively unexplored field which we will study.

2.2 Redefining the Problem

Building on this foundation, our project aims to extend the MobileNetv2 approach with more complex data. Firstly, we want to reproduce the baseline training strategy of Demirel[1], in an attempt to reach the same accuracy under the same conditions. From there, we wish to evaluate the performance on a dataset that contains mushrooms that display high visual similarity while also lowering the total amount of training images and, finally, we wish to assess how accuracy, precision and recall change as the number of species in the dataset increases.

3 Methods

3.1 Dataset

We opted for another dataset, than the one used in the research paper: “Mushroom Species Recognition” by Leonardo Cofone (<https://www.kaggle.com/datasets/zlatan599/mushroom1/data>). A dataset containing over 100 species of mushrooms captured in natural environments varying in lighting conditions, backgrounds, and viewing angles. We chose four mushrooms similar to the ones used by Demirel to recreate his results. Due to a vast imbalance in the class distribution we also chose to take a subset of 200 images of each species in our own experiments, and from here we split the dataset with 80% of the data to be used for training and 20% used for validation (see Appendix A to see the class distribution).

3.2 Data Preprocessing and Augmentation

All images were preprocessed using the standard MobileNetV2 input pipeline. This includes resizing to 224×224 and normalization using ImageNet statistics. To reduce overfitting and improve the model’s generalization performance, data augmentation was applied on-the-fly to the training set only. The following transformations were used: Random rotation (range 10% range), width shift (range 10% range), height shift (10% range) and zoom (10% range) degrees. These augmentation techniques simulate the natural viewing conditions in which mushrooms may be encountered in real environments, such as variations in angle, scale, and perspective.

3.3 Network Architecture

The final model is built on the MobileNetV2 architecture, which was originally designed for efficient image classification on resource-constrained devices. MobileNetV2 is pretrained on ImageNet, which

makes it a convenient foundation for our mushroom classification task, as the model has already learned a rich set of visual features from over 14 million images.

As shown in the diagram describing the architecture (see Appendix C), the network begins with an initial convolutional layer, followed by a sequence of inverted residual bottleneck blocks. Each block expands the feature channels with a 1×1 convolution, applies a depthwise 3×3 convolution, and then projects the features back to a lower dimension. Finally, a skip connection is used to preserve information and support stable training. After the final bottleneck block, MobileNetV2 applies a 1×1 convolution followed by global average pooling to produce a 1280-dimensional feature vector. Once we start fine-tuning in phase 3, an additional dropout layer is inserted before the final fully connected classifier in an attempt to reduce overfitting on the mushroom dataset. The original classification head is replaced with a new dense layer that outputs twelve classes, corresponding to the expanded set of mushroom categories.

3.4 Experiments

All experiments were based on the MobileNetV2 architecture and followed the same preprocessing pipeline. The motivation of the experimental design was not simply to maximise accuracy, but to systematically explore how the model behaves as task complexity increases. The experiments were structured as a controlled progression in difficulty, where Phase 1 represents the baseline, Phase 2 introduces increased visual similarity, and Phase 3 increases both visual similarity and class diversity. Adjustments to the training process were only introduced when the model clearly was struggling to follow the increase in complexity. While this introduces a minor limitation in terms of direct comparability with the earlier phases, the adjustments are considered a part of investigating the model's scalability and practical limitations in a more realistic fashion.

3.4.1 Phase 1: Baseline experiment (four visually distinct species)

Phase 1 was conducted with the intention of creating a true baseline using four visually distinct mushroom species. The standard MobileNetV2 architecture and default training configuration were applied without any additional modifications. Under these simple and controlled conditions, the model achieved an overall good performance in detection. This result established a foundation for model performance under ideal and visual differences and served as the comparison point for all subsequent experiments.

3.4.2 Phase 2: Visual similarity experiments (four visually similar species)

With the intention of stress testing our model, the dataset was replaced with a collection of four visually similar mushroom species, while all architectural and training parameters were kept identical to the baseline setup. When applied to this more challenging dataset, the model displayed clear signs of inconsistencies during training, including overshooting and a substantial decrease in

validation accuracy. To address this issue while preserving as much comparability as possible with Phase 1, only the learning rate was adjusted.

3.4.3 Phase 3: Scalability experiment (twelve species)

In Phase 3, the complexity of the task was further increased by expanding the dataset to include twelve species. This came from the motivation of exploring the scalability and examining the real world applications of our model. When the original and slightly adjusted training configuration from the previous phases was applied, the model performed poorly and struggled to perform at an acceptable level. This behaviour indicated that the baseline architecture and training setup were no longer sufficient to handle the increased complexity. In an attempt to make the model reach the same performance as in phase 1, several new targeted modifications were introduced.

Adjusted hyperparameters: We first reduced the batch size to 16, which immediately improved stability and accuracy. The number of epochs was increased to 30 to smooth the loss and accuracy curves. **Stronger data augmentation:** To handle the twelve-class dataset, augmentation was expanded with horizontal flips and brightness variation, increasing visual diversity and improving generalization. **Learning-rate scheduling:** A step-decay schedule was introduced to reduce instability observed at higher learning rates. Starting from 0.001, the learning rate was multiplied by 0.9 each epoch, which produced more consistent convergence. **Full-network fine-tuning:** All MobileNetV2 layers were unfrozen and trained using cosine decay with a lower initial learning rate ($1e-4$). This allowed the backbone to adapt more effectively to mushroom-specific features. **Secondary fine-tuning:** A final refinement stage was performed for seven epochs using cosine decay with an even smaller learning rate ($1e-6$) to ensure smooth convergence. **Additional regularization:** Two dropout layers (0.6 and 0.4) were added to reduce overfitting, followed by another short fine-tuning cycle with cosine decay ($1e-6$). These adjustments produced a more stable and better-generalizing model for the expanded and visually similar twelve-class dataset.

4 Evaluation

Each experiment was evaluated using validation accuracy/loss, precision, recall, F1-score, confusion matrices, and training curves. Presenting the three phases together highlights how MobileNetV2's[4] performance changes as the classification task increases in difficulty, from visually distinct species, to visually similar species, to a larger varied 12-class dataset.

4.1 Training stability and accuracy

Our baseline model, replicating the experiments done by Demirel[1] we also managed to archive a validation accuracy of 96.91% and a 0.09 loss. It already had an accuracy of above 90% after the first epoch showing that the model had an easy time distinguishing between the visually distinct

mushrooms.

When we switched to the visually similar, using the exact same model we first saw an issue of overshooting, as seen in Figure 1, with validation accuracy of 58.75% and a 0.21 loss. We figured this was due to the smaller dataset, so we lowered the learning rate to account for this which resulted in a much nicer training/validation curve (Figure 1), and a new result of 89.38% validation accuracy with a 0.30 loss. Thus, still a pretty good model, but significantly poorer than the baseline model.

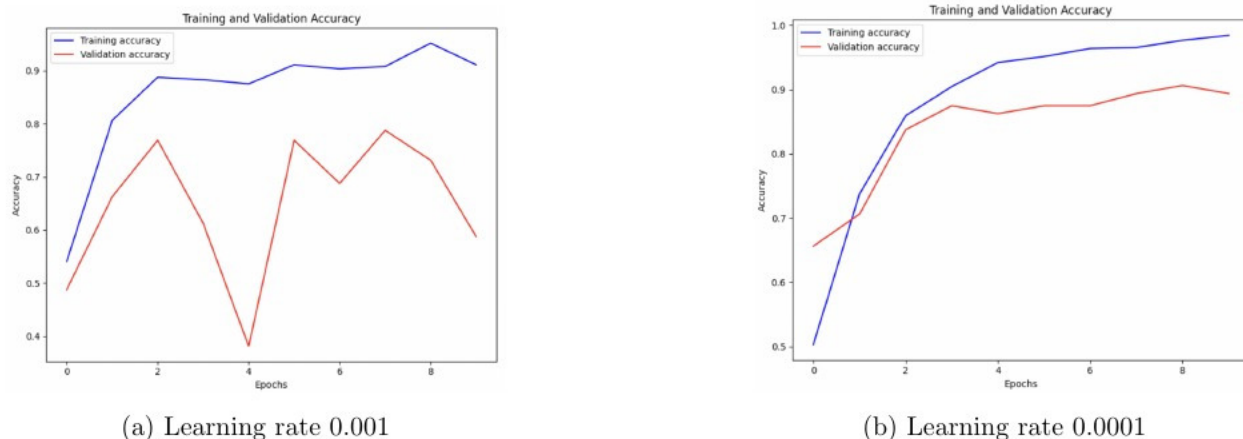


Figure 1: Training and validation accuracy for two learning rates.

Finally, we had the dataset with a larger number of species. When running the dataset using the baseline model with the reduced learning rate, we got a validation accuracy of 85.62% with a 0.56 loss. Again, we see the model having increased difficulty as expected, but with the improvements explained in the methods section, we did manage to get it to 92.32% accuracy with a 0.33 loss as seen in Figure 2.

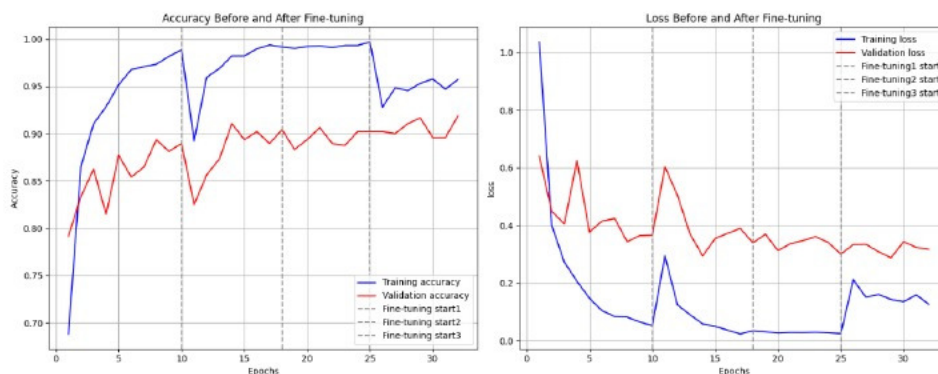


Figure 2: Training and validation rates final

4.2 Classification performance

For the visually distinct species, the model achieved uniformly high precision, recall, and F1-scores (0.96–0.99). *Amanita muscaria* reached near-perfect performance ($F1 = 0.99$), and even the smallest class, *Psilocybe pelliculosa*, maintained an F1-score of 0.96, showing that class imbalance was not an issue at this difficulty level.

For the visually similar *Amanita* species, all classes experienced a noticeable drop, with F1-scores ranging from 0.87–0.94 and a macro F1 of 0.90. *Amanita muscaria* remained the best performer ($F1 = 0.94$), while *A. flavoconia* and *A. persicina* showed slightly lower precision and recall, reflecting the challenge of distinguishing between species with subtle morphological differences.

For the twelve-class dataset, several species still achieved very high scores. *Laetiporus sulphureus* and *Psilocybe pelliculosa* both reached $F1 = 0.99$ with perfect recall, and *Amanita muscaria* maintained a strong F1 of 0.97. The lowest performance was seen in *Boletus edulis* ($F1 = 0.84$), primarily due to reduced precision. Moderate scores were also observed for the two *Daedaleopsis* species ($F1 = 0.88$ – 0.90), which aligns with their subtle visual overlap.

4.3 Confusion matrix

For the distinct four-class dataset, the matrix shows strong diagonal dominance with almost no off-diagonal errors, reflecting that the model found these species easy to separate (see subsection B.1 for confusion matrix). In the visually similar dataset, the remaining confusions occurred primarily between *Amanita flavoconia*, *Amanita persicina*, and *Amanita augusta*, all of which share similar cap and colour patterns (see subsection B.2 for confusion matrix).

The twelve-class confusion matrix shows the most varied structure but remains strongly diagonal overall. Misclassifications is mainly among visually overlapping species, for example, *Boletus edulis* receives false positives from *Suillus grevillei* and *Leucoagaricus americanus*, and the two *Daedaleopsis* species show light cross-confusion. However, species with distinctive morphology such as *Laetiporus sulphureus*, *Psilocybe pelliculosa*, and *Amanita muscaria* has strong diagonals with almost no misclassifications.

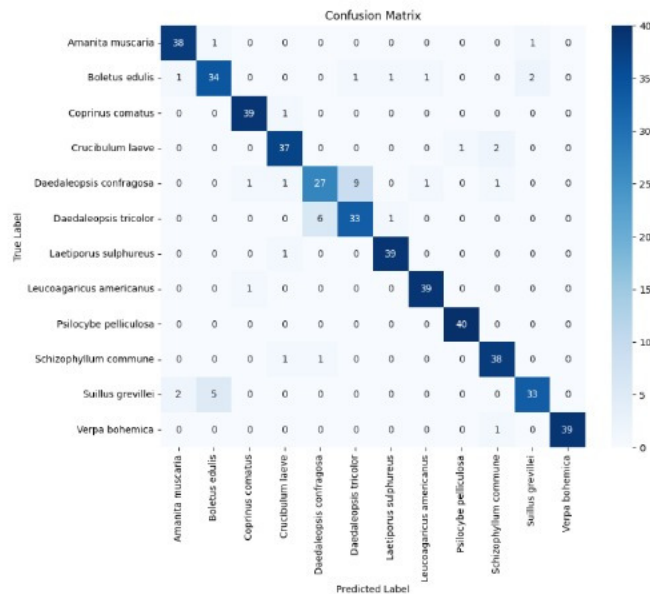


Figure 3: Confusion matrix phase 3

5 Limitations

To get a better understanding of the model’s weaknesses, a set of the ten least confident misclassified validation images was examined. These samples illustrate the kinds of visual features that lead to incorrect predictions. For each image, both the predicted label and the true label are shown and also how confident the model was when classifying these. The misclassified examples typically include mushrooms photographed in challenging conditions, such as low light, unusual angles, other objects disturbing the view of the mushroom, or with significant background clutter. Several errors also occur among visually similar species, where subtle differences in cap structure, coloration, or pore patterns are difficult for the model to distinguish. By inspecting these failures, it becomes clear that the model tends to rely on overall appearance rather than fine morphological details, making it vulnerable when species share similar global features. To be fair, the misclassified images always predict a species that is similar to the true label.



Figure 4: Lack of mushrooms

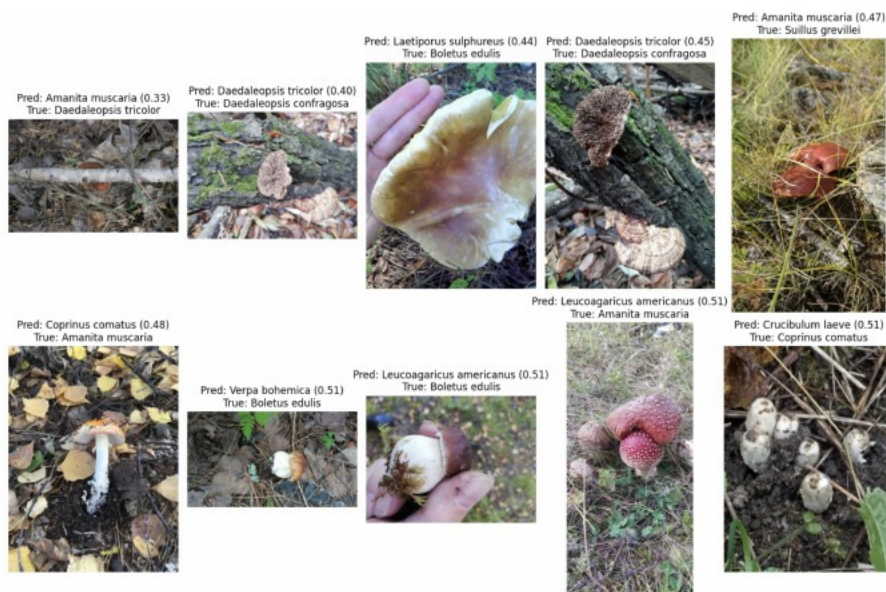


Figure 5: Failed classifications

These examples highlight the importance of larger and more diverse training data, as well as more advanced feature extraction methods, for improving robustness in mushroom classification. In addition to the ten misclassified samples from the validation set, a few particularly problematic images were also identified in the training dataset. These images contain only leaves or dirt and (hardly) no visible mushroom at all, making it near impossible for the model to extract meaningful

fungus features. As expected, the model produced an incorrect prediction, not because of subtle species similarities but because these images lack any relevant information.

6 Discussion

The purpose of this study was to examine whether the MobileNetV2 used in the referenced article can maintain high performance when scaled from simple mushroom classification tasks to increasingly complex challenges. Across the three experimental phases, our findings show a clear pattern: while MobileNetV2 performs extremely well in visually distinct settings, its performance degrades as visual similarity and class diversity increase. This reflects both the strengths and limitations of lightweight CNNs when applied to real world biological classification and does not naturally capture the subtle cues required to separate species with overlapping colours, shapes, and textures.

Misclassification analysis supports this interpretation. The ten worst predictions typically involved images taken in unusual lighting with strong shadows, low contrast, or obstructions such as surrounding vegetation. They also included species whose visual characteristics vary significantly depending on maturity or environmental conditions. These examples reveal that the model relies heavily on coarse visual cues such as overall color and cap texture rather than fine structural details. This behavior is expected for lightweight CNNs and mirrors findings from related studies, but it limits the model's reliability in realistic field conditions where lighting and context are unpredictable.

Another important challenge was the sensitivity of the model to training hyperparameters. Although a high learning rate worked in the simplest scenario, it caused severe instability in more complex settings. Only after reducing the learning rate, applying stronger data augmentation, introducing additional dropout, and fine-tuning the entire backbone did the model achieve satisfactory performance on the twelve-class phase.

Overall, our findings suggest that MobileNetV2 is a strong baseline for mushroom classification but not robust enough for real-world identification of visually similar species, especially in safety-critical contexts (e.g. falsely classifying a poisonous mushroom for an edible mushroom). Performance improvements achieved through augmentation, learning-rate scheduling, and full-model fine-tuning show that the architecture has potential, but its reliance on coarse features and its difficulty handling subtle variations remain huge limiting factors. Future work could focus on making the dataset larger (and wider), architectures designed for fine-grained recognition (such as Attention-based CNNs or Metric-learning models e.g. Siamese or Triplet networks), and methods that incorporate additional visual or contextual information. Despite its limitations, this project demonstrates that lightweight CNNs can perform well across a range of classification difficulties, while also revealing the practical challenges that arise when applying them to complex and similar mushroom species.

References

- [1] Y. Demirel and G. Demirel, “Deep learning based approach for classification of mushrooms,” *Gazi University Journal of Science Part A: Engineering and Innovation*, vol. 10, no. 4, pp. 487–498, 2023.
- [2] N. Zahan, M. Z. Hasan, M. A. Malek, and S. S. Reya, “A deep learning-based approach for edible, inedible and poisonous mushroom classification,” in *2021 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD)*, pp. 440–444, 2021.
- [3] M. J. Y. Sutayco and M. V. C. Caya, “Identification of medicinal mushrooms using computer vision and convolutional neural network,” in *2022 6th International Conference on Electrical, Telecommunication and Computer Engineering (ELTICOM)*, pp. 167–171, 2022.
- [4] Keras Team, “Mobilenet (keras applications).” <https://keras.io/api/applications/mobilenet/>, 2024. Accessed: 2025-02-11.

7 Appendix

A Class distribution

Baseline	
Species	Dataset size
Coprinus comatus	1615
Amanita muscaria	4576
Laetiporus sulphureus	2590
Psilocybe pelliculosa	292
Visually similar species	
Species	Dataset size
Amanita muscaria	200
Amanita flavoconia	200
Amanita rubescens	200
Amanita amerirubescens	200
Large number of species	
Species	Dataset size
Amanita muscaria	200
Boletus edulis	200
Coprinus comatus	200
Crubibulum laeve	200
Daedaleopsis confragosa	200
Daedaleopsis tricolor	200
Laetiporus sulphureus	200
Leucoagaricus americanus	200
Psilocybe pelliculosa	200
Schizophyllum commune	200
Suillus grevillei	200
Verpa bohemica	200

B Confusion Matrices

B.1 Phase 1

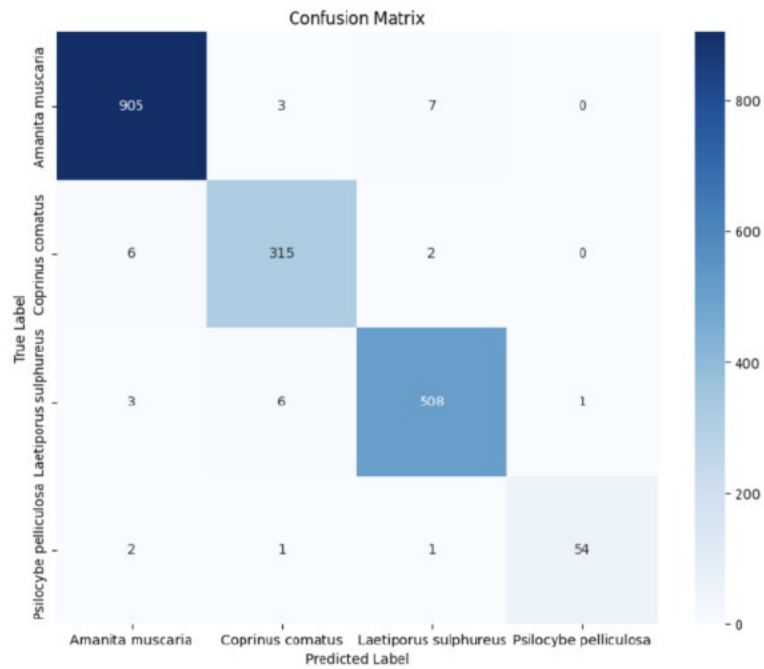


Figure 6: Confusion Matrix Phase 1

B.2 Phase 2

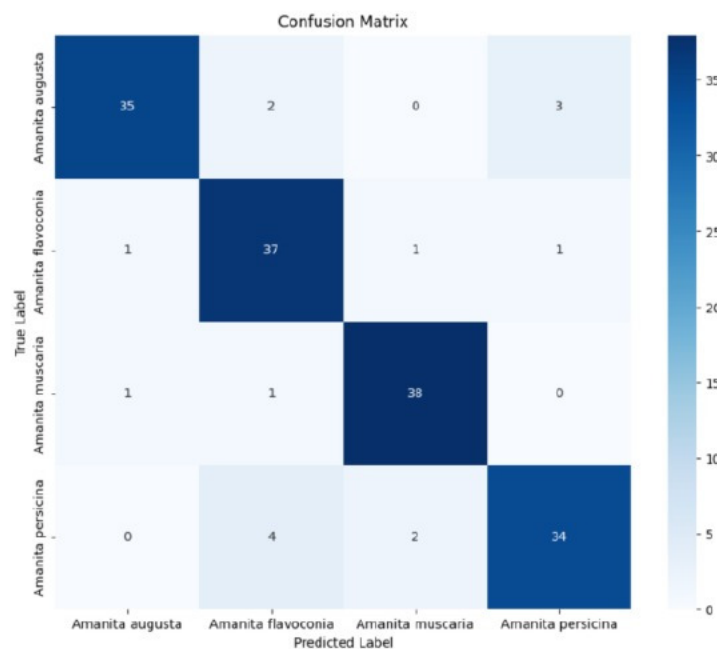


Figure 7: Confusion Matrix Phase 2

C Network Architecture

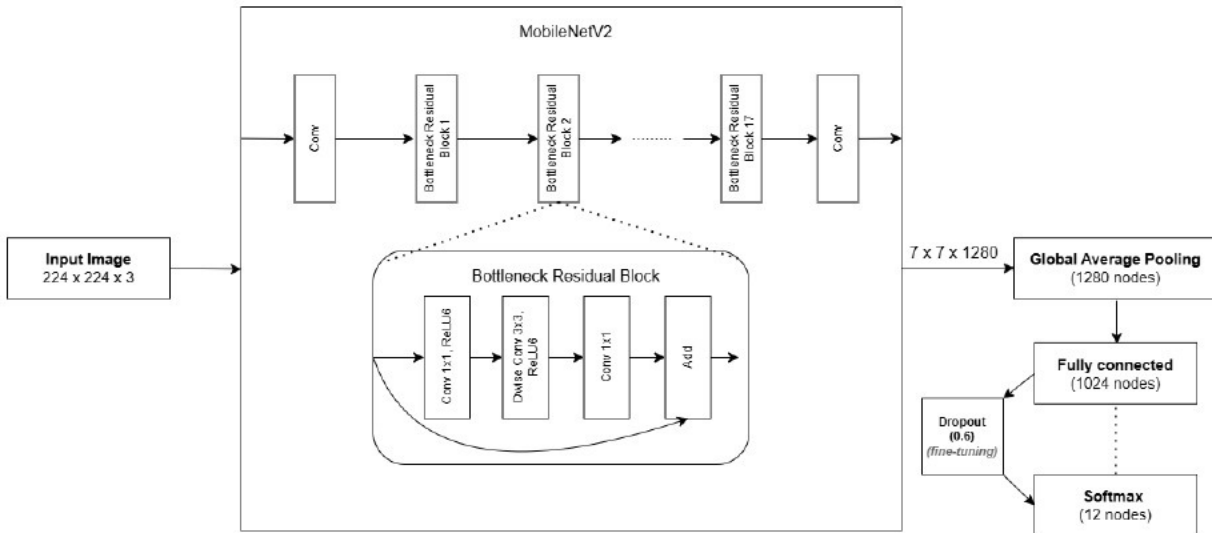


Figure 8: Network architecture